

Cryptographic Protocol Analysis and Verification (v8)

Problem Mentor(s):

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Description: The design of cryptographic protocols to provide key distribution and agreement, authentication and secrecy goals is a difficult problem. As the world has become more dependent upon computer networks for the transfer of sensitive information and financial transactions, it is more important than ever that the cryptographic protocols that protect the information and transactions be correct. Unfortunately, many of the cryptographic protocols that have been designed over the years have not provided the intended security properties or only provided those properties under a restrictive set of assumptions that are not longer valid in the current environment where the protocol is deployed. The problems are often the result of the way the protocols are designed, relying on the experience of the designer, their unexpressed assumptions, and their knowledge of security protocol failures. To address this issue, a significant amount of research has been dedicated to the development of tools and techniques to formally verify the security properties of cryptographic protocol specifications against well-defined assumptions. There are a number of tools that have been developed over the years to verify security properties. NSA has supported the development of two such tools, the Cryptographic Protocol Shapes Analyzer and the Maude-NPA (NRL Protocol Analyzer). A number of other tools exist, such as Tamarin, ProVerif, and others. Given the large number of cryptographic protocols that are designed, it is imperative that we validate the security properties of those that are designed and introduce the tools into the design process to eliminate flawed standards being introduced.

Possible Project Directions:

- There are a variety of ways that this project can be taken. One would be to choose a standard cryptographic protocol and analyze it with one of the tools to verify the protocols security properties or identify security flaws. If you identify flaws in a standard security protocol, that would be a publishable result. If on the other hand, you are able to verify the security of a protocol undergoing consideration for standardization, the standards body would be interested in the report. You may choose any security protocol you find interesting.
- There is increased interest in the use of agentic AI and the security issues surrounding it. For example, a protocol like the Model Context Protocol (MCP) still refers to the use of standard protocol security properties such as authentication, confidentiality, and authorization and relies on traditional cryptographic security protocols TLS and OAuth without defining how to appropriately apply the protocols with there many options to provide the appropriate security. Formal analysis of these traditional protocols for use in agentic AI is something that can be pursued with the traditional formal analysis tools. Another approach is to understand agentic protocols at a higher level in terms of the communication. As LLMs are capable of generating their own protocols for communicating with tools and other LLMs, there are security risks associated with the alignment of LLMs to the original goals and intent. Well known attacks such as prompt injections can now be made through the tools an LLM interacts. Current tools are unable to address these higher level attacks. Are there approaches that can be proposed? Is it possible to formalize such high level protocols to allow proofs of meaningful security goals?
- One of the difficulties with using verification tools is the manner in which specifications are written. In general, protocol specifications are not written in an easy to analyze or easy to implement format. The IETF is considering approaches to make use of formal language in IETF standards to make protocols more secure by design and remove ambiguity in implementation. One approach to this project would be to explore languages for protocol specifications that would be more formal and support a bridge to tools such as CPSA, Maude-NPA and others. This of course would entail understanding the different approaches to modeling the protocols that the tools take. A domain specific language for protocol specification with formal semantics may be a way to achieve this goal. As an example, one could look at Cryptol, a domain specific language and tools to specify and verify cryptography. Information on Cryptol can be found at <https://cryptol.net/>.
- An entirely different take on the project would be to determine if an implementation is consistent with the specification. The tools verify high-level models of a specification and even if the specification is correct, implementation errors may lead to security vulnerabilities. Developing approaches to abstract the protocol from the implementation and verify it is consistent with the specifications could be another approach.
- These are suggestions, feel free to propose other research to improve the security of network protocols.

Type of Project: Applied Research, Basic Research

Relevant References for Problem:

1. CPSA can be found at: <https://hackage.haskell.org/package/cpsa> or <https://github.com/mitre/cpsa>
2. Maude-NPA can be found at: <http://maude.cs.uiuc.edu/tools/Maude-NPA/>
3. Tamarin Prover can be found at: <https://tamarin-prover.github.io>
4. ProVerif can be found at: <https://bblanche.gitlabpages.inria.fr/proverif/>
5. Example reports of a formal analysis of a protocol with CPSA can be found at: <https://arxiv.org/abs/2003.07421> and <https://eprint.iacr.org/2025/2067>
6. Security Protocols for space systems can be found at <https://public.ccsds.org/default.aspx>

Required Skills for Carrying Out Project:

- Network protocols
- Basic cryptography, programming

Recommended Skills for Carrying Out Project:

- Network security

Level of Suitable Students: Undergraduate, Masters, PhD

Length of Project: Two semesters possible (e.g., capstone project)

Other Expectations:

- Weekly progress reports submitted to the INSuRE Hub. An initial meeting at the start of the project in addition to the mid-semester progress presentation and final project presentation. Additional meetings can be arranged at the request of the team.
- Depending upon the project you choose to pursue, I can point you at additional resources. Analyzing a protocol would be within the realm of a semester project. If choosing to analyze a protocol, from my experience, CPSA is the easier of the various tools available to learn and training in the use of CPSA is available from the UMBC Protocol Analysis Lab.

